L3a: Equity Exchanges, Growth Rates, and Stylized Facts

CHEME 5660 Cornell University

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Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we connect equity-market infrastructure to price data, convert prices into growth rates and returns, and use recurring empirical patterns to test a model of equity behavior.

Today's concepts

- Describe equity claims: distinguish the corporation, an exchange, a stock index, and an exchange-traded fund.
- Compute growth rates: calculate continuously compounded growth rates and convert them to log returns using $r = (\Delta t)g$.
- Test return models: use return distributions and autocorrelation to check whether a model matches observed market data.

Companion notebook: L3a lecture notebook

Lectures and Examples

Lecture:
CHEME-5660-L3a-Lecture-Equity-Exchanges-Return-StylizedFacts-Fall-2026.ipynb

Example: CHEME-5660-L3a-SVD-SP500-Example-Fall-2026.ipynb

Use SVD to identify common patterns in the growth rates of S&P 500 constituents.

Example: CHEME-5660-L3a-StylizedFacts-Example-Fall-2026.ipynb

Compare distribution fits, inspect tail-index stability, and contrast the memory of signed growth with the memory of growth magnitude.

Link: https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git

Exchanges, Stocks, and ETFs

Before measuring an equity return, separate the marketplace from the securities traded there. These objects interact, but they are not interchangeable.

- An exchange is an organized venue that matches orders and imposes listing and trading rules.
- A stock is a residual ownership claim on a corporation.
- An index is a rule-based numerical portfolio benchmark. It is not itself a security.
- An exchange-traded fund, or ETF, is a security whose portfolio may seek to track an index and whose shares trade during the day.

Today: a market price is the price available at a particular time and trading venue.

Exchanges Connect Buyers and Sellers

An exchange brings orders together and supports price discovery, the process by which trading establishes a market price, and liquidity, the ability to trade without moving that price excessively.

- Several U.S. venues compete for the same order, including NYSE, Nasdaq, Cboe BZX, and IEX.
- A broker may route one order to a venue or divide it among venues, subject to available prices, liquidity, and the order instructions.
- After execution, separate clearing and settlement infrastructure completes the transfer of securities and funds.

The National Best Bid and Offer, or NBBO, reports the highest displayed bid and the lowest displayed ask across the national market system.

Order Books and the NBBO

Order books and the NBBO

National Best Bid and Offer

NASDAQ • XYZ

SELL / ASK	SIZE	PRICE
	200	50.07
	400	50.05
BEST	100	**50.04**

spread = $0.03

BUY / BID	SIZE	PRICE
BEST	300	**50.01**
	500	50.00
	200	49.99

TOP OF BOOK • XYZ

VENUE	BID	ASK	SPREAD
NYSE	50.00	50.05	0.05
Nasdaq	50.01	50.04	0.03
Cboe BZX	50.02	50.06	0.04
IEX	50.01	50.06	0.05

best bid and best ask sit on different venues

NBBO • XYZ

BEST BID	BEST ASK
$50.02	**$50.04**
Cboe BZX	Nasdaq

spread $0.02

tighter than any single venue’s own spread

Stocks Are Ownership Claims

A stock, also called a share or equity security, represents fractional ownership in a corporation. It is a residual claim, not a promise of fixed repayment.

What the shareholder may receive

- Voting rights and cash distributions such as dividends.
- A claim on remaining assets only after contractual creditors are paid.
- A capital gain if another investor later pays a higher share price.

Limited liability ordinarily caps the shareholder's loss at the invested amount. It does not protect the share price from falling to zero.

Source: U.S. Securities and Exchange Commission, Stocks

ETFs Turn Portfolios into Tradable Securities

An ETF pools assets into a portfolio and issues shares that trade on an exchange throughout the day. Each share represents an interest in that portfolio.

- A broad-market ETF can provide diversified exposure, but sector, leveraged, inverse, and single-stock ETFs may be highly concentrated.
- The net asset value, or NAV, is the fund's assets minus its liabilities, divided by shares outstanding.
- An ETF has both a market price and a NAV per share. They are usually close, but the ETF can trade at a premium or discount to NAV.

Like a stock, an ETF produces an observable sequence of market prices.

From Securities to Price Data

Let $S_t > 0$ denote the observed market price of a stock or ETF at time t. The statistic computed from that sequence depends on what the sequence means.

Two data conventions

- A price return uses the quoted price series and omits cash distributions such as dividends.
- A total return includes distributions, commonly through an adjusted-price series whose vendor convention must be documented.

State the adjustment rule, sampling interval, and time units before comparing growth rates or risk statistics.

Reward

We measure price reward with the continuously compounded growth rate, denoted by g. Let $S_{j-1}^{(i)} > 0$ and $S_j^{(i)} > 0$ be consecutive prices for firm i, separated by $\Delta t_j > 0$ years. The price evolution is:

$$S_j^{(i)} = S_{j-1}^{(i)} \exp\left(g_{j,j-1}^{(i)} \Delta t_j\right).$$

Solving for the rate gives:

$$g_{j,j-1}^{(i)} = \underbrace{\frac{1}{\Delta t_j} \log\left(\frac{S_j^{(i)}}{S_{j-1}^{(i)}}\right)}_{\text{growth rate}}.$$

The growth rate has units of inverse years. Its sign records whether the price increased, decreased, or remained unchanged. The notebook uses closing prices, so this is a price-growth measure unless an adjusted series is selected.

Growth Rates and Log Returns

Let g_t be the continuously compounded growth rate over an interval of length $\Delta t_t > 0$ years. The corresponding log return r_t is dimensionless and is given by:

$$\underbrace{r_t = (\Delta t_t) g_t}_{\text{log return}} = \log\left(\frac{S_t}{S_{t-1}}\right).$$

Log returns add across adjacent intervals. For positive prices $S_0, S_1, \ldots, S_T$, the cumulative log return is:

$$\sum_{t=1}^{T} r_t = \underbrace{\sum_{t=1}^{T} (\Delta t_t) g_t}_{\text{time-weighted growth}} = \log\left(\frac{S_T}{S_0}\right).$$

Use g to compare annualized rates. Convert to r before aggregating across time.

Excess Growth Compares an Asset with a Benchmark

Let g_t be the equity growth rate and let $g_{f,t}$ be a declared benchmark growth rate over the same interval. The excess growth rate and excess log return are given by:

$$\underbrace{\bar{g}_t = g_t - g_{f,t},}_{\text{excess growth}} \qquad \underbrace{\bar{r}_t = (\Delta t_t)\bar{g}_t}_{\text{excess log return}} \;.$$

- $\bar{g}_t > 0$: the equity price grew faster than the benchmark.
- $\bar{g}_t < 0$: the equity price grew more slowly than the benchmark.
- $\bar{g}_t = 0$: both produced the same continuously compounded growth rate over the interval.

Careful: the benchmark must use the same currency, horizon, and time units. A positive excess rate is not a guarantee of positive wealth.

Algorithm

Let $\mathcal{D}_i = \{S_1^{(i)}, \dots, S_T^{(i)}\}$ be a positive price series for firm i, let $\Delta t > 0$ be the interval in years, and let g_f be the benchmark growth rate.

1. For $t = 2, \dots, T$, look up $S_{t-1}^{(i)}$ and $S_t^{(i)}$.
2. Compute the excess growth observation:

$$\bar{g}_t^{(i)} \leftarrow \underbrace{\frac{1}{\Delta t} \log\left(\frac{S_t^{(i)}}{S_{t-1}^{(i)}}\right)}_{g_t^{(i)}} - g_f.$$

3. Store $\bar{g}_t^{(i)}$. The T prices produce $T - 1$ observations.

Set $g_f = 0$ when the required output is the growth rate rather than the excess growth rate.

Common Patterns Across Many Assets

Let $G \in \mathbb{R}^{m \times n}$ collect centered growth rates for m dates and n assets. Its singular value decomposition is:

$$G = U\Sigma V^{\top} = \sum_{j=1}^{\text{rank}(G)} \underbrace{\sigma_j \mathbf{u}_j \mathbf{v}_j^{\top}}_{\text{rank-one pattern}}, \qquad \sigma_1 \geq \sigma_2 \geq \cdots \geq 0.$$

Large singular values identify common patterns that account for much of the variation across assets.

SVD finds patterns in the data; it does not explain their economic causes. Results depend on centering, scaling, missing-data treatment, and the selected assets.

Notebook: CHEME-5660-L3a-SVD-SP500-Example-Fall-2026.ipynb

Risk

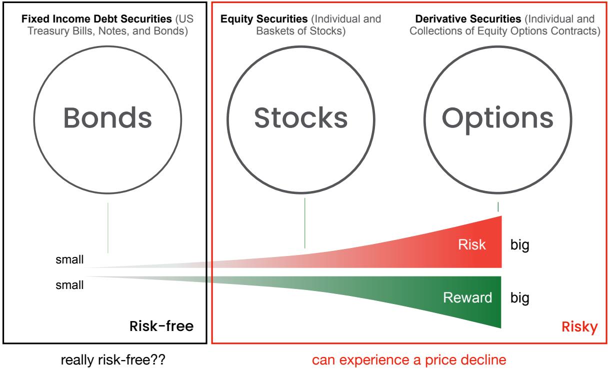
Risk is uncertainty about loss or gain. A common dispersion measure is the sample standard deviation of the growth rate, denoted by σ_g.

Suppose T prices produce the $T-1$ observations $g_2, g_3, \dots, g_T$. Let g' be their sample mean. The sample volatility is:

$$g' = \frac{1}{T-1}\sum_{t=2}^{T} g_t, \qquad \sigma_g = \underbrace{\sqrt{\frac{1}{T-2}\sum_{t=2}^{T}(g_t - g')^2}}_{\text{growth-rate volatility}}.$$

The statistic has the same inverse-year units as g. It measures dispersion around the sample mean, not tail probability, drawdown, liquidity, or the magnitude of a specific loss.

Return and Risk Measure Different Things



Assets with similar sample means and standard deviations can still have very different tails, drawdowns, dependence, and liquidity.

Other Risk Measures

Choose a risk measure that matches the type of loss relevant to the decision.

- Value at Risk: a loss threshold exceeded with a stated small probability over a stated horizon.
- Expected Shortfall: the average loss beyond that threshold.
- Maximum drawdown: the largest peak-to-trough decline in value.
- Downside deviation: dispersion measured only below a declared target.
- Beta: sensitivity to a selected market benchmark.
- Tracking error: volatility of the return difference from a benchmark.

Ask: what type of loss matters, over what horizon, and will this statistic detect it?

Stylized Facts of Equity Growth Rates

A stylized fact is a recurring empirical pattern observed across many assets or markets. It is not an exact law for every ticker and sample. It is a pattern that a useful model should be able to reproduce.

- Heavy tails: large movements occur more often than a fitted Gaussian model predicts.
- Weak linear autocorrelation: signed growth rates often have little serial linear dependence at ordinary liquid-market horizons.
- Volatility clustering: large movement magnitudes occur near other large movements, and calm observations near other calm observations.

Sources: Cont (2001); Mandelbrot (1963)

Heavy Tailed Return Distributions

A power-law tail decays more slowly than an exponential tail. Let X be a random variable, let $\alpha > 0$ be its tail index, and let $L(x)$ be slowly varying. A two-sided power-law tail satisfies:

$$\mathbb{P}(|X| > x) \sim \underbrace{L(x)x^{-\alpha}}_{\text{tail probability}} \qquad \text{as } x \to \infty.$$

A smaller α means slower decay and a heavier tail. The classical strict thresholds identify moments that fail to exist for a pure power-law tail:

- $\alpha < 1$: the mean fails to exist.
- $\alpha < 2$: the variance fails to exist.
- $\alpha < 4$: the fourth moment fails to exist.

Careful: at the boundary values 1, 2, and 4, moment existence depends on finer tail detail. Heavy-tailed does not always mean power-law.

Hill's Estimator of the Tail Index

Hill's estimator applies to positive observations. For signed growth rates, analyze one tail or the magnitude series $|g|$. Let $X_{(1)} \geq X_{(2)} \geq \cdots \geq X_{(n)} > 0$ be descending order statistics, and let $k < n$ select the upper-order observations. The estimate is:

$$\hat{\alpha}_k = \underbrace{\left[\frac{1}{k}\sum_{i=1}^{k}\log\left(\frac{X_{(i)}}{X_{(k+1)}}\right)\right]^{-1}}_{\text{estimated tail index}}.$$

The cutoff k balances bias and variance. One arbitrary cutoff is not a diagnostic: compute $\hat{\alpha}_k$ over a range of k values and look for a stable region.

Companion example: the stylized-facts notebook uses a Hill stability plot and treats $k = \lfloor\sqrt{n}\rfloor$ as a reproducible reference, not a uniquely correct cutoff.

Autocorrelation

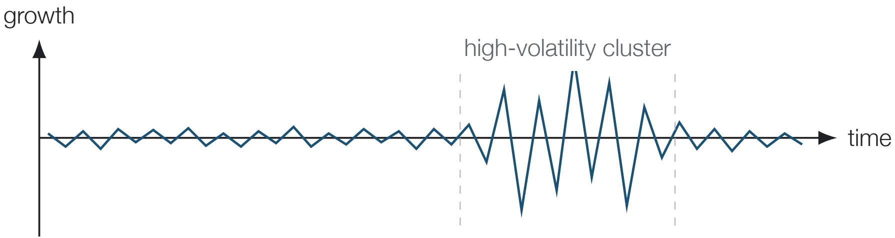
Let $g_2, \ldots, g_T$ be growth-rate observations with sample mean g', and let k be a lag measured in sampling intervals. Define the empirical autocovariance and autocorrelation by:

$$\widehat{R}(k) = \frac{1}{T-1}\sum_{t=2}^{T-k}(g_t - g')(g_{t+k} - g'), \qquad \underbrace{\rho(k) = \frac{\widehat{R}(k)}{\widehat{R}(0)}}_{\text{autocorrelation}}.$$

The normalization gives $\rho(0) = 1$. For many equity samples, $\rho(k)$ is near zero at nonzero lags: past signed growth provides little linear help in predicting future signed growth.

Near-zero autocorrelation does not establish independence. Dependence can remain in growth magnitude, trading microstructure, or other nonlinear functions of the series.

Volatility Clustering



Volatility clustering means large movements tend to occur near other large movements, and small movements near other small movements. The cluster contains both signs, so the dependence concerns magnitude rather than a persistent direction.

Diagnosing Volatility Clustering

Let $g_2^{(i)}, \ldots, g_T^{(i)}$ be the growth-rate series for firm i, and let $\overline{(g^2)_i}$ be the mean squared growth rate. The empirical autocorrelation of squared growth rates is:

$$\hat{\rho}_i(k) = \frac{\displaystyle\sum_{t=2}^{T-k} \left[(g_t^{(i)})^2 - \overline{(g^2)_i}\right] \left[(g_{t+k}^{(i)})^2 - \overline{(g^2)_i}\right]}{\displaystyle\sum_{t=2}^{T} \left[(g_t^{(i)})^2 - \overline{(g^2)_i}\right]^2}.$$

Volatility clustering appears when $\hat{\rho}_i(k) > 0$ across many lags and fades slowly toward zero. A model can therefore match weak signed-growth ACF and still fail because it assumes fixed, independent volatility.

Why it matters: conditional volatility changes risk forecasts, option values, hedge ratios, and capital allocation.

Example

Diagnosing Stylized Facts in Daily Equity Growth Rates

The companion notebook turns each stylized fact into a model check:

- Compare empirical growth rates with fitted Normal, Laplace, and Student-t densities.
- Inspect a Hill stability plot instead of reporting one cutoff.
- Compare the ACF of signed growth with the ACF of squared growth.
- Translate the diagnostics into requirements for the next model.

Notebook: CHEME-5660-L3a-StylizedFacts-Example-Fall-2026.ipynb

Optional Advanced Material

Advanced 1: advanced/order_book/
CHEME-5660-L3a-Advanced-LimitOrderBook-Mechanics-Fall-2026.ipynb

Advanced 2: advanced/market_impact/
CHEME-5660-L3a-Advanced-Market-Impact-Models-Fall-2026.ipynb

Advanced 3: advanced/stylized_facts/
CHEME-5660-L3a-Advanced-StylizedFacts-Fall-2026.ipynb

Optional, not prerequisites for L3b. The first develops the mechanics of a price-time-priority limit-order book; the second distinguishes mechanical execution cost from linear, order-flow-imbalance, and transient-impact models; the third deepens the stylized-facts diagnostics and hands them off to L6.

Notebooks: lectures/week-3/L3a/advanced, also linked from the lecture notebook.

Summary

This lecture introduced equity securities, continuously compounded growth rates, and recurring empirical patterns in financial return data.

Key takeaways

- **Market infrastructure and securities are different.** Exchanges discover prices; stocks and ETFs are claims traded there.
- **Return statistics require conventions.** The growth rate annualizes and the log return aggregates. State price versus total return.
- **Stylized facts are model requirements.** Heavy tails and clustered volatility expose failures of a fixed-variance independent model.

These diagnostics become tests for the models that follow.

That’s a wrap. What are some of the interesting things we discussed today?